

Quiz 4
CSE471: System Analysis and Design
SET-D

Name:

ID:

Total Marks: _____ / 20

Time: 25 minutes

(CO2)

Q1. You should show the proper units for all the questions, detailed calculations, and/or formulas.

Attributes	Average Size (bytes)
Name	11
Id	7
Address	3
SSN	5
Email	9
Designation	4
Mobile	11
Salary	6

- a. Find the **Record Size** in **bits**. [2]

$$\begin{aligned}\text{Record Size} &= \text{Summation of all the attribute's average size} \\ &= ((11 + 7 + 3 + 5 + 9 + 4 + 11 + 6) * 8) \text{ bits} \\ &= 448 \text{ bits}\end{aligned}$$

- b. The system's overhead will be 0.25. Calculate the **Total Record Size** in **bits**. [2]

$$\begin{aligned}\text{Total Record Size} &= \text{Record Size} + (\text{Record Size} * \text{overhead}) \\ &= 448 + (448 * 0.25) \text{ bits} \\ &= 560 \text{ bits}\end{aligned}$$

- c. The initial table size is 45,000. Calculate the **Initial Table Volume** in **bits**. [2]

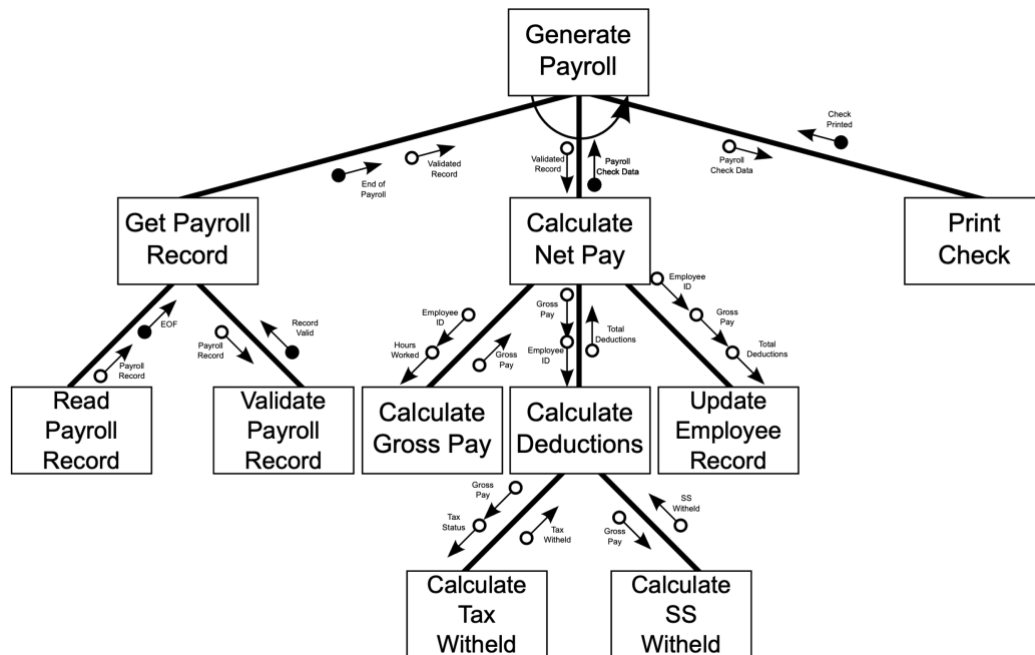
$$\begin{aligned}\text{Initial Table Volume} &= \text{Initial Table Size} * \text{Total Record Size} \\ &= 45000 * 560 \text{ bits} \\ &= 25,200,000 \text{ bits}\end{aligned}$$

- d. Calculate the **Final Table Volume** in **bits** after 48 months; quarterly growth is 2500. [4]

$$\begin{aligned}\text{Final Table Volume} &= \text{Initial Table Volume} + (\text{Growth per unit} * \text{Total Record Size} * \text{total time}) \\ &= (25,200,000 + ((2500/3) * 560 * 48)) \text{ bits} \\ &= 47,600,000 \text{ bits}\end{aligned}$$

(CO2)

Q2.



Generate Payroll is numbered as 1.0.

- How many subordinate modules are there for Calculate Net Pay? [2]
Ans: 3
- What kind of module is 1.3? [2]
Ans: Subordinate
- Which data couples are sent down by module 1.2? [2]
Ans: Employee ID, Hours Worked, Gross Pay, Total Deductions
- What are the fan-in and fan-out numbers for 1.2.2? [2]
Ans: Fan-in: 1, Fan-out: 2
- What type of cohesion Update Employee Record is in? [2]
Ans: Functional